

Hamza Fayaz

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI Engineer building production-grade systems in agentic AI, computer vision, and backend automation. Experienced in turning manual operational workflows into scalable pipelines across order operations, content workflows, and edge AI prototypes. Skilled in Python, LangGraph/LangChain, FastAPI, and cloud deployment.

EDUCATION

University of Engineering and Technology Mardan | 3.3 CGPA

Bachelor of Science in Computer Software Engineering

EXPERIENCE

EARL OF ETON LIMITED

Apr 2025 to Present

AI Automation Engineer (Remote)

Lead automation initiatives across customer operations, order workflows, and digital content processes. Work cross-functionally with operations and media teams replacing manual tasks with scalable systems, integrating AI-driven workflows, CRM processes, API-based automation, WhatsApp Business API, and Meta API integrations for social publishing

Harp Technologies

Dec 2025 to Mar 2026

AI/ML Engineer (Contract, In person)

Contributed to a cross-functional AI-powered dual-screen display program focused on edge inference and visual optimization. Led model research and benchmarking for image enhancement and depth estimation on Hailo NPU, developed a prototype 2D-to-3D glasses-free display pipeline, and implemented eye-tracking logic for dynamic depth alignment.

PROJECTS

Linnworks Supervised Agentic Automation + Van-Sales Mobile Workflow (AWS, Step Functions, FastAPI, React Native/Expo)

- Built a deterministic order automation system over Linnworks and email with LLM-assisted classification, drafting, and a van-sales mobile app into one end-to-end flow.
- Automated customer contact through email, reply handling, and Human-in-the-Loop approval for risky order actions.

whatsapp-mcp-assistant (FastMCP, Go, whatsmeow, Python)

- Built a production WhatsApp automation server using FastMCP and a Go whatsmeow bridge, exposing real WhatsApp sessions as LLM-callable tools inside Cursor.
- Enforced Human-in-the-Loop by blocking all outbound messages unless explicitly approved at runtime.

Jarvis: Autonomous Agentic AI PC Voice Assistant (FYP)

- Built a mini-Jarvis assistant with full voice-based control of the PC, including reading/explaining context, writing outputs, and executing multi-step tasks with memory.
- Engineered offline-first voice pipeline and LangGraph multi-agent orchestration using LM Studio + Kaggle GPU offloading.

Agentic RAG Sub-Agents: Multi-Agent Document Intelligence System (Python, pgvector, supabase, React, LangSmith)

- Built a production-grade RAG system with hybrid search, Reciprocal Rank Fusion, and Cohere reranking using raw Python SDK no framework abstractions.
- Implemented pgvector semantic search, metadata filtering, and LangSmith observability for retrieval evaluation

SKILLS

Languages: Python, SQL, JavaScript, C/C++ , GO

Automation: n8n, Make (Integromat), Zapier, Airtable, Workflow Orchestration, API Integration, Cursor, Web Scraping

Frontend & Mobile: Electron, React Native (Expo Go)

AI/ML & LLM: LangChain, LangGraph, LlamaIndex, RAG, Hugging Face, Claude (Anthropic), Prompt Engineering, MCP Server, A2A

Computer Vision: OpenCV, YOLO, MediaPipe, Image Segmentation, Object Detection, Depth Estimation

Backend & APIs: FastAPI, Flask, REST APIs, Authentication, API Integration

Data & Vector Stores: PostgreSQL, MySQL, SQLite, MongoDB, Pinecone, Chroma, Weaviate, Supabase

Cloud & DevOps: Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, AWS, Azure, GCP, Linux, Git, Bedrock, Self-hosted n8n